Haoyu Ren, PhD Candidate

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PROFILE

Ph.D. candidate in Mathematics and Statistics with experience in biostatistics, clinical trial methodology, causal inference, statistical modeling, machine learning, Bayesian and real-world data analysis. Experienced in collaborating with clinicians and cross-functional research teams, developing simulation studies, analyzing clinical and real-world healthcare data, and communicating statistical results for oncology, biomarker, and patient outcomes research. Proficient in R, SAS, and statistical programming, with strong interest in innovative statistical methods for clinical development and drug/vaccine research.

EDUCATION

University of Maryland, Baltimore County, Baltimore, MD

M.S. in Mathematics and Statistics, 2024

Ph.D. in Mathematics and Statistics, GPA: 3.8/4.0 | September 2021 – Present

Expected graduation: 2026

Advisors: Dr. Yi Huang and Dr. Chixiang Chen

University of Maryland, College Park, College Park, MD

B.S. in Mathematics and Statistics, cum laude, GPA: 3.9/4.0

EXPERIENCE

Biostatistics Intern | June 2026 – August 2026

Biostatistics and Research Decision Sciences, Merck, Upper Gwynedd, PA

- Evaluated how scan timing affected median progression-free survival (PFS) bias, Cox hazard ratio estimates, and log-rank test power under constant and delayed treatment effects.
- Quantified the trade-off between median PFS bias and the number of scans per patient to inform when trials could shift to less frequent imaging.

Research Assistant | June 2024 – Present

Division of Biostatistics and Bioinformatics, Department of Epidemiology and Public Health, University of Maryland, School of Medicine

Collaborating with University of Maryland, School of Medicine

- Collaborated with clinicians and cross-functional research teams to translate clinical questions into appropriate statistical models, conduct data analyses, and interpret study findings.

Causal Machine Learning

- Co-led the development of a new doubly robust method for continuous treatments while addressing missing data using a change-point model.
- Co-led the development of a new method for analyzing heterogeneous treatment effects under continuous treatment while addressing missing data using a varying-coefficient model.

Causal Meta-Analysis

- Co-led the development of a new weighting function with smaller bias and variance under rare events than existing weighting functions.

Teaching Assistant | September 2021 – May 2024

Department of Statistics, University of Maryland, Baltimore County

- Explained mathematical and statistical concepts to students.
- Held office hours and created and graded quizzes and homework assignments.

PUBLICATIONS

- **H. Ren**, T. Wen, J. Falvey, Y. Huang, and C. Chen, "*A Doubly Robust Approach to Evaluating the Impact of Hospital Length of Stay on Post-Fracture Recovery in Older Adults With Cognitive Impairment*," *Statistics in Medicine* 45, no. 23-24 (2026): e70768, https://doi.org/10.1002/sim.70768.
- Shen, B., **Ren, H.,** Shardell, M., Falvey, J., & Chen, C. (2023). *Analyzing risk factors for post-acute recovery in older adults with Alzheimer's disease and related dementia: A new semiparametric model for large-scale Medicare claims.* Statistics in Medicine, 43(5), 1003–1018. https://doi.org/10.1002/sim.9982

  Contribution: *Coded and evaluated the method through simulation.*

- Falvey, J. R., Chen, C., …, **Ren, H.,** … (2023). *Associations of days spent at home before hip fracture with postfracture days at home and 1-year mortality among Medicare beneficiaries living with Alzheimer's disease or related dementias.* The Journals of Gerontology, Series A: Biological Sciences and Medical Sciences, 78(12), 2356–2362. https://doi.org/10.1093/gerona/glad158
  Contribution: *Applied clustering methods to analyze multiple longitudinal outcomes.*
- Kardon, A., Kim, D., **Ren, H.,** … (2024). *A propensity-score weighted analysis of short-term corticosteroid therapy for refractory pain following spontaneous subarachnoid hemorrhage.* Neurocritical Care.
  Contribution: *Applied propensity-score weighting to compare treatment effects.*
- Graeme, F., Chen, C., …, **Ren, H.,** … (2025). *Microbubble-enhanced focused ultrasound with temozolomide for patients with high-grade gliomas: A multicenter, open-label, phase 1/2 trial.* Lancet Oncology.
  Contribution: *Analyzed cfDNA and fragment length ratio data.*
- Kardon, A., **Ren, H.,** … (2025). *Early markers of systemic inflammation are not related to pain burden after aneurysmal subarachnoid hemorrhage: A multicenter observational study.* Neurocritical Care. DOI: 10.1007/s12028-025-02396-w.
  Contribution: *Conducted variable selection and tested the proportional odds assumption for ordinal logistic regression.*
- Fahim, A., …, **Ren, H.,** … (2025). *Systemic Cyst(e)inase Administration Induces Ferroptosis and Synergizes with Temozolomide in Glioblastoma.* iScience. https://doi.org/10.1016/j.isci.2025.114350
- Blair, R., …, **Ren, H.,** … (2026). *Post-transcriptional control of HIF-1α by MBNL1 restrains hypoxia-driven stemness in GBM.* Neoplasia. https://doi.org/10.1016/j.neo.2025.101262
  Contribution: *Applied a linear mixed-effects model to analyze the performance of three genes across different cell types and conditions.*

MANUSCRIPTS READY TO SUBMIT

- **Ren, H.,** …, Chen, C. *A Novel Causal Machine Learning Approach to Study Threshold Effects of Continuous Exposure Using Multi-Source Linked Medicare Claims Data.* Prepared for submission to Biometrics.
- Harrison, D. M., …, **Ren, H.**, Chen, C., … (2026). *Cellular-level Quantification of the Impact of Multiple Sclerosis on Retinal Ganglion Cell Bodies and Axons Using Adaptive Optics–Optical Coherence Tomography.* Manuscript in preparation.
  Contribution: *Compared retinal cell and RNFL densities across MS subgroups and evaluated inter-rater reliability and clinical associations.*
- Falvey, J., …, **Ren, H.**, … (2026). *Engaging Community and Municipal Stakeholders to Improve High-Quality Aging in Place after Hip Fracture.* Manuscript in preparation.
  Contribution: *Performed XGBoost variable selection, model comparison, and odds-ratio analysis.*
- Morris, N., …, **Ren, H.**, … (2026). *The Relationship Between Blood and Edema Clearance with Pain Burden after Aneurysmal Subarachnoid Hemorrhage: A Retrospective Cohort Study.* Manuscript in preparation.
  Contribution: *Evaluated associations between CT hemorrhage-score changes and pain outcomes using adjusted and sensitivity analyses with G-formula.*
- Wen, T., …, **Ren, H.**, … (2026). *Hospital Characteristics and Recovery after Hip Fracture in the United States.* Manuscript in preparation.
  Contribution: *Developed a risk-adjusted XGBoost/SHAP ensemble to rank hospital characteristics and predict days at home.*

AWARDS

- Dean's List, Fall 2020 and Spring 2020, University of Maryland, College Park
- Graduated cum laude, University of Maryland, College Park

PRESENTATIONS

- ENAR, New Orleans, March 2025.
  Contributed Paper: Breaking Barriers in Statistical Insights for Health Disparities and Outcomes
  A User-Friendly Approach to Detecting the Heterogeneity of Hospital Length of Stay Effects on Post-Acute Recovery Outcomes for Older Adults with Cognitive Conditions
- JSM, Boston, August 2026.
  Contributed Paper: Section on Statistics in Epidemiology — Causal Inference Methods and Applications in Epidemiology
  A Novel Causal Machine Learning Approach to Study Threshold Effects of Continuous Exposure
- ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop, Maryland, 2026.
  Poster
  A Novel Causal Machine Learning Approach to Study Threshold Effects of Continuous Exposure

SKILLS

- **Programming and Software:** MATLAB, SAS, R, Java, C++, Minitab
- **Other:** Word, Excel, PowerPoint, cross-functional collaboration, statistical modeling, study design, missing data, longitudinal data analysis, Bayesian, real-world data analysis, categorical data analysis, machine learning, profile search, mixed model, multivariate analysis, HPC.